

Lecture 9 (10/5)

Bias-variance trade off, confidence intervals and least squares

Bias and consistency

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$$E[\hat{\theta}] \rightarrow \theta \quad \text{and} \quad \text{se}(\hat{\theta}) \rightarrow 0 \quad \text{as } N \rightarrow \infty.$$

Example

For $Y_i \sim \text{Normal}(\mu, \sigma^2)$, is $\hat{\mu}_1 = (Y_1 + Y_2)/2$ biased? Is it consistent? Find its sample distribution.

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For $Y_i \sim \text{Normal}(\mu, \sigma^2)$, is $\hat{\mu}_2 = \bar{Y} + 1/2$ biased? Is it consistent? Find its sample distribution.

Example

Suppose $Y_i \sim \text{Normal}(\mu, \sigma^2)$ and μ is known to us

$$\hat{\sigma}_\mu^2 = \frac{1}{n} \sum_i (Y_i - \mu)^2$$

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Example

Suppose $Y_i \sim \text{Normal}(\mu, \sigma^2)$ and μ is NOT known to us

$$\hat{\sigma}_0^2 = \frac{1}{n} \sum_i (Y_i - \bar{Y})^2$$

Is this an unbiased estimator?

True vs. estimated sample distribution

For $Y_i \sim \text{Normal}(\mu, \sigma^2)$, the *true* sample distribution of $\hat{\mu} = \bar{Y}$ is

$$\hat{\mu} \sim \text{Normal}\left(\mu, \frac{\sigma^2}{N}\right).$$

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But we don't know μ or σ ! In practice we plug in our estimates to get the estimated sample distribution

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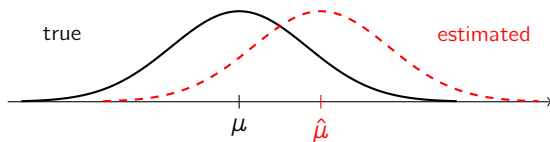
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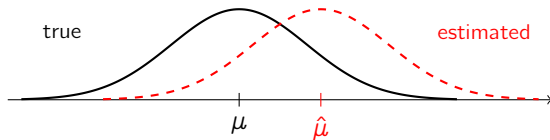
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We need to estimate the parameter in the sample distribution, which means the entire distribution is itself a random variable. Still, its width tells us how far μ might plausibly be from $\hat{\mu}$. This idea is formalized with confidence intervals.

Confidence intervals

The 95% confidence interval:

$$\left[\hat{\theta} - 1.96 \text{se}(\hat{\theta}), \quad \hat{\theta} + 1.96 \text{se}(\hat{\theta}) \right].$$

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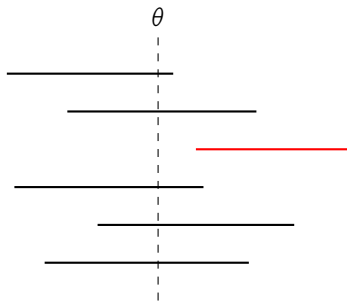
For an approximately Normal sample distribution, 95% of values fall within 1.96 standard errors of the center.

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- **Correct:** if we generate many replicates of the data, the true θ falls in the resulting CI for 95% of them.
- **Not correct:** there's a 95% chance θ lies in *this particular* interval.



Each bar is the 95%-CI from one replicate; about 95% of them cover θ (the red one misses).

Example

Suppose $\sigma = 1$ is known. How many samples are needed so the 95%-CI for μ has margin of error less than 1?

The least squares estimator

Data $(X_1, Y_1), \dots, (X_N, Y_N)$ from $Y | X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\epsilon^2)$. How do we estimate β_1 ?

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Replace the covariance and variance with their sample versions:

$$\hat{\beta}_1 = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_i (X_i - \bar{X})^2}, \quad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}.$$

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This is the least squares estimator, since

$$(\hat{\beta}_0, \hat{\beta}_1) = \underset{b_0, b_1}{\operatorname{argmin}} \sum_i (Y_i - b_0 - b_1 X_i)^2,$$

i.e. it minimizes the sum of squared residuals.